proj

persistence promote...

Question !!

both traits. As yet, however, it is unknown whether persistence is associated with increased mutation rates and how this could affect the likelihood for genetic resistance to evolve.

Does bacterial persistence facilitate the evolution of antibiotic resistance by providing a reservoir of viable cells and increasing mutation rates?

background

Hypotheses

(i.) persistence increases likelihood of resistance not only by increasing viable cells during treatment, but also due to pleiotropic increased in mutation rates

(ii.) Mutants w/ high persistence & high mutation rates may have selective advantage & accelerate evolution of resistance

E: Why include immunity?

↳ clinical relevance?

makes setup more complicated though

bacPersistSupp.pdf

Model (of within-host population dynamics)

Normal N, Persister P cells

Wild Type W, 3-step resistant mutant strains ($R1$, $R2$, $R3$)

Innate immune effector cell E

Antibiotic concentration $A(t)$

ODEs:

N_W, P_W, N_{R1}, P_{R1}, N_{R2}, P_{R2}, N_{R3}, P_{R3}, E, A

equation system:

$$N_W'(t) = g_W(A)\,N_W(t) - a\,N_W(t) + b\,P_W(t) - \mu_W\,N_W(t) - \lambda_N\,N_W(t)\,E(t)$$
$$P_W'(t) = a\,N_W(t) - b\,P_W(t) - \mu_P\,P_W(t) - \lambda_P\,P_W(t)\,E(t)$$
$$N_{R1}'(t) = g_{R1}(A)\,N_{R1}(t) - a\,N_{R1}(t) + b\,P_{R1}(t) + \mu_W\,N_W(t) - \mu_{R1}\,N_{R1}(t) - \lambda_N\,N_{R1}(t)\,E(t)$$
$$P_{R1}'(t) = a\,N_{R1}(t) - b\,P_{R1}(t) + \mu_P\,P_W(t) - \mu_P\,P_{R1}(t) - \lambda_P\,P_{R1}(t)\,E(t)$$
$$N_{R2}'(t) = g_{R2}(A)\,N_{R2}(t) - a\,N_{R2}(t) + b\,P_{R2}(t) + \mu_{R1}\,N_{R1}(t) - \mu_{R2}\,N_{R2}(t) - \lambda_N\,N_{R2}(t)\,E(t)$$
$$P_{R2}'(t) = a\,N_{R2}(t) - b\,P_{R2}(t) + \mu_P\,P_{R1}(t) - \mu_P\,P_{R2}(t) - \lambda_P\,P_{R2}(t)\,E(t)$$
$$N_{R3}'(t) = g_{R3}(A)\,N_{R3}(t) - a\,N_{R3}(t) + b\,P_{R3}(t) + \mu_{R2}\,N_{R2}(t) - \lambda_N\,N_{R3}(t)\,E(t)$$
$$P_{R3}'(t) = a\,N_{R3}(t) - b\,P_{R3}(t) + \mu_P\,P_{R2}(t) - \lambda_P\,P_{R3}(t)\,E(t)$$
$$E'(t) = q\,(e_{max} - E(t)) - l\,E(t)$$

Simulate:

(i.) effect of varying persistence/mutation rate on

(i.) time to resistance emergence

(ii.) selective advantage of rare mutants

(iii.) pop.-dynamics under treatment

Resistance Evolution:

$WT \longrightarrow R1 \longrightarrow R2 \longrightarrow R3$

MIC increases →

Why 3-step resistance?

↳ partial resistance mutations need to survive long enough to acquire additional mutations.

↳ Persisters ensure that survival window

Normal N: growing, susceptible to Ax

Persister P: dormant, tolerant to Ax, less sensitive to immune

$a(t)$ rate of normal → persister

(depends on pop.-size)

↳ more crowding → more switching

b constant rate persister → normal

Selective advantage of mutants with pleiotropically linked persistence and mutation rates

Invasion fitness, *i.e.* the long-term relative fitness of a mutant relative to the wild-type, under conditions described above, was calculated as:

$$\frac{P_{R1}(t_2) \times (1 - P_W(t_1))}{P_W(t_1) \times (1 - P_{R1}(t_1))}$$

observation / question

that the link between persistence and resistance is not merely based on the increased number of cells available for mutation, but that persistence could also be directly linked with altered mutation rates and increased probabilities of mutating into an antibiotic-resistant genotype. First, both mutation and persistence have been suggested to be induced

→ is persistence directly linked w/ altering mutation rates & increased prob. of mutating Ax-resistant genotype?

Answer: Yes-ish (experiments show overlap)

mutating into an antibiotic-resistant genotype. First, both mutation and persistence have been suggested to be induced by the activity of different stress responses [23–25]. Hence, high persister fractions and high mutation rates could coincide in stressful environments due to a partial overlap in underlying mechanisms [17]. Second, both high-persistence

underlying mechanisms [17]. Second, both high-persistence and mutator phenotypes are known to thrive in fluctuating environments and upon sudden environmental changes [11,

(prior observations - experimental)

↓

both traits. As yet, however, it is unknown whether persistence is associated with increased mutation rates and how this could affect the likelihood for genetic resistance to evolve.

Question!

Does bacterial persistence promote evolution by increasing mutation rates?
If so, how?

↓

Importantly, tolerance and persistence have also been suggested to have the potential to act as a stepping stone towards the evolution of genetic resistance, as persisters provide a viable cell reservoir from which resistant mutants can emerge by horizontal gene transfer or de novo chromosomal mutations [16, 17]. This idea was first put forward

hypothesis!
(possible mechanism)

persisters provide viable cell reservoir that "incubates" resistance evolution via HGT & mutations